

Part I

Parameters of Leavitt Law



1 Observables - Luminosity, Reddening and Distance

1.1 Why Does a Star Shine?

The luminosity of a star originates from nuclear fusion reactions occurring in its core. During star formation, gravitational collapse of interstellar gas leads to extremely high central pressures and densities, raising the core temperature to several million kelvin. Under these conditions, thermal energies become sufficient to overcome the Coulomb repulsion between protons, enabling nuclear fusion.

In the Sun and other low-mass main-sequence stars, energy production is dominated by the proton–proton (pp) chain, in which hydrogen nuclei fuse to form helium through a sequence of reactions discussed in the following chapter. Approximately 7×10^8 metric tons of hydrogen are converted into helium each second, releasing energy according to mass–energy equivalence.

A fraction of this energy is carried away by neutrinos, which escape the stellar interior almost unimpeded, while the remainder is transported outward by radiation and convection and ultimately emitted from the stellar surface as electromagnetic radiation. This radiation spans a broad range of wavelengths, with the peak emission in the optical band for solar-type stars, and propagates through space as starlight.

1.2 Luminosity

Light is electromagnetic (EM) radiation that propagates in vacuum at a constant speed of $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$. The human eye is sensitive only to a narrow portion of the EM spectrum—the optical band—spanning approximately 350–700 nm in wavelength.

⁰Carina Nebula (NGC-3324) Credit: James Webb Space Telescope First Images. NASA, ESA, CSA, and the STScI Webb ERO Production Team (2022)

This range represents a small subset of the full electromagnetic spectrum, which extends from high-energy gamma rays to long-wavelength radio emission.

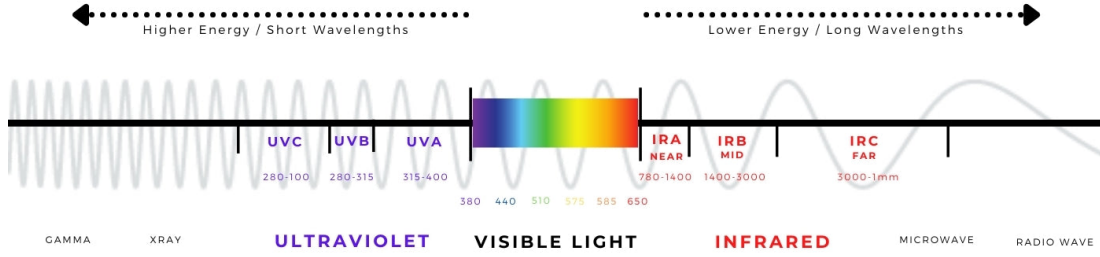


Figure 1.1: Spectrum of Light showing different bands with respective wavelength range. Source: www.healthlighting.com

The spectral distribution of radiation emitted by matter is governed by its thermal state. An idealized emitter, known as a blackbody, absorbs all incident radiation and re-emits it with a spectral energy distribution (SED) determined solely by its temperature. With increasing thermal excitation, the emission peak shifts toward shorter wavelengths, while the total radiative output increases. This behavior is quantitatively described by Planck's law, which underpins modern radiative theory.

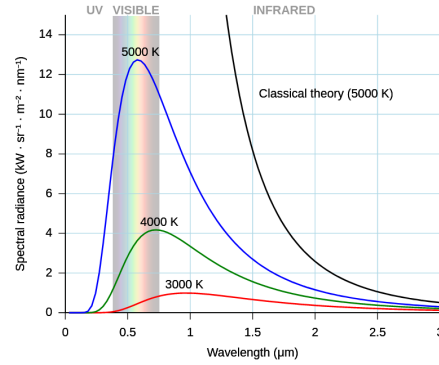


Figure 1.2: *Black body radiation curves based on Planck's Law for various temperatures. The maxima of the curves shift according to Wien's displacement law. Source: Wikipedia commons*

According to Planck (1900), the monochromatic specific intensity of blackbody radiation describes

the radiative power emitted per unit frequency (or wavelength), per unit area, per unit solid angle, by a blackbody in thermal equilibrium. It depends only on the frequency (or wavelength) and the temperature of the emitter and is given by Planck's law,

$$B_{\lambda}(T) = \frac{2hc^2}{\lambda^5} \left(e^{\frac{hc}{\lambda kT}} - 1 \right)^{-1} \quad (1.1)$$

where λ is the wavelength, T the temperature, h Planck's constant, and k Boltzmann's constant and B_{λ} is the specific intensity in units of $\text{W m}^{-2} \text{sr}^{-1} \text{m}^{-1}$, .

Integrating it over the outward hemisphere above the blackbody surface gives emergent monochromatic flux F_{λ}

$$F_{\lambda}(T) = \int B_{\lambda}(T) \cos \theta \, d\Omega = \pi B_{\lambda}(T). \quad (1.2)$$

The total radiative flux is obtained by integrating the spectral flux density over all wavelengths,

$$F = \int_0^{\infty} F_{\lambda} \, d\lambda = \sigma T^4, \quad (1.3)$$

where σ is the Stefan–Boltzmann constant.

Consider a star of radius R emitting a surface flux F_{λ} . At a distance d from the star, the observed monochromatic flux is reduced according to the inverse-square law,

$$f_{\lambda} = \left(\frac{R}{d}\right)^2 F_{\lambda}. \quad (1.4)$$

A detector measures radiation over a finite wavelength range defined by its instrumental response (or sensitivity) function E_{λ} . The detected signal is therefore

$$s = \int_0^{\infty} f_{\lambda} E_{\lambda} \, d\lambda. \quad (1.5)$$

In observational astronomy, this integrated flux is recorded within discrete wavelength intervals, referred to as *photometric bands*, and is commonly expressed in logarithmic units known as *magnitudes*.

1.2.1 Magnitude Scale

The ancient Greek astronomer Hipparchus (c. 190 – c. 120 BCE) compiled a star catalogue and introduced a scale of stellar brightness in units of magnitude (mag), where 1 mag corresponds to the brightest stars visible to the unaided eye, and 6 mag corresponds to the dimmest. The magnitude scale was revisited by Pogson in the 1850s [Pogson \(1856\)](#) and later calibrated by Zöllner in 1861 using a visual photometer [Zöllner \(1861\)](#). In their work, comparisons were made between artificial and real stars using two Nicol prisms to establish a more precise magnitude scale.

According to Pogson’s definition, the difference in apparent magnitudes between two stars is proportional to the logarithm of their radiation flux ratio (s_1/s_2), i.e.,

$$m_1 - m_2 = 2.5 \log \left(\frac{s_1}{s_2} \right). \quad (1.6)$$

Thus, a difference of 5 magnitudes corresponds to a brightness ratio of 100. For ex-

ample, α Lyr (Vega) has a visual magnitude of 0.14 mag, while α Cyg (Deneb) has 1.33 mag, making it fainter than Vega. The Sun, Moon, Venus, Mars, Jupiter, and three stars—Sirius, Canopus, and α Centauri—are the only celestial objects with negative apparent magnitudes, placing them among the brightest objects in the night sky.

1.2.2 Color Index ($m_1 - m_2$)

The difference between stellar magnitudes measured in two distinct wavelength bands is termed the *color index*. Because stellar flux varies with wavelength, the measured magnitude of a star depends on the observational filter employed, leading to observable color differences among stars.

As an illustration, Betelgeuse appears brighter than Sirius when observed through a red filter, despite Sirius being the brightest star in the visual band. This contrast reflects the differing spectral energy distributions of the two stars: Sirius emits most strongly at shorter wavelengths, whereas Betelgeuse radiates predominantly at longer wavelengths. Such variations arise from differences in their photospheric effective temperatures.

The physical basis for interpreting stellar colors was established by Karl Schwarzschild, who demonstrated that the shape of a stellar spectral energy distribution is primarily governed by temperature. His analysis showed that color measurements provide a quantitative means of estimating stellar photospheric temperatures (Schwarzschild, 1900).

Building upon this theoretical framework, W. W. Morgan and H. L. Johnson developed a standardized photometric system to measure stellar colors and magnitudes reliably. The Johnson–Morgan UBV system consists of three broadband filters with effective wavelengths in the ultraviolet ($\lambda_{\text{eff}} = 365$ nm), blue ($\lambda_{\text{eff}} = 440$ nm), and visual ($\lambda_{\text{eff}} = 548$ nm) regions (Johnson and Morgan, 1953). This system was subsequently extended into the red and infrared through the addition of further passbands, commonly specified in micrometers (μm):

Band	<i>U</i>	<i>B</i>	<i>V</i>	<i>R</i>	<i>I</i>	<i>J</i>	<i>H</i>	<i>K</i>	<i>L</i>	<i>M</i>	<i>N</i>	<i>Q</i>
λ_{eff} (μm)	0.36	0.44	0.55	0.70	0.90	1.25	1.63	2.20	3.60	5.00	10.6	21

By combining broadband photometry with spectral classification, Morgan further developed the MK system, which allows stars to be classified simultaneously by effective temperature (spectral type) and luminosity class. These advances established color

indices as a fundamental observational tool in stellar astrophysics, providing a quantitative link between stellar color and physical properties.

The stellar energy distribution can be approximated by Planck's radiation law. Adopting Wien's approximation ($c_2/(\lambda T) \gg 1$), the monochromatic radiative flux is given by

$$F_\lambda \propto \exp\left(-\frac{c_2}{\lambda T_{\text{eff}}}\right), \quad (1.7)$$

where $c_2 = hc/k_B$ is the second radiation constant equivalent to $= 1.4388 \text{ cm K}$.

Corresponding magnitude would be,

$$m_\lambda = -2.5 \log F_\lambda + \text{const.} \quad (1.8)$$

Assuming that the detector sensitivity peaks near the effective wavelengths of the photometric filters, the relation between the color index $B - V$ ($m_B - m_V$) and the stellar effective temperature can be expressed as

$$B - V = \frac{2.5 c_2 \log e}{T_{\text{eff}}} \left(\frac{1}{\lambda_B} - \frac{1}{\lambda_V} \right) + \text{const.} \quad (1.9)$$

Using $c_2 = 0.014 \text{ m K}$, $\lambda_B = 440 \text{ nm}$, and $\lambda_V = 548 \text{ nm}$, this reduces to

$$B - V \approx \frac{0.7 \times 10^4}{T_{\text{eff}}} + \text{const.} \quad (1.10)$$

This expression clearly demonstrates that the color index $B - V$ provides a direct estimate of a star's effective temperature, highlighting the utility of photometric measurements in stellar astrophysics.

Adopting $\alpha \text{ Lyr}$ (Vega; spectral type A0V) as the standard star, the photometric zero points in the U , B , and V bands are defined such that

$$U - B = B - V = 0, \quad (1.11)$$

implying $C_{UB} = C_{BV} = 0$.

For convenience, the color index $U - B$ is denoted by C_{UB} . Stars with $C_{BV} \lesssim 0.5$ are classified as blue, while those with $C_{BV} \gtrsim 1.5$ are considered red.

1.2.3 Absolute Magnitude (M)

When a luminous object is observed from different distances, it appears to have varying brightness because flux decreases with increasing distance. So, how can we compare the intrinsic brightness of two remote stars? To enable standardized comparisons, astronomers adopt a reference distance of 10 parsecs (pc). The radiant flux of a star measured at this distance defines its *absolute magnitude*, denoted as M_λ . The subscript λ indicates the specific photometric band (filter) used in the observation, as brightness varies across different bands.

Given this framework, if m_λ represents the apparent magnitude of a star observed at a distance d (in parsecs), and M_λ is its absolute magnitude, the relationship between them is given by the following formula:

$$M_\lambda = m_\lambda - 2.5 \log \left(\frac{d[\text{pc}]}{10} \right)^2$$
$$M_\lambda = m_\lambda - 5 \log d[\text{pc}] + 5 \quad (1.12)$$

The above is a simple relation which relates the observable luminosity with the distance of any stellar object.

Note 1.2.1: True Absolute Magnitude

The above relation must be corrected for the effect of interstellar extinction, A_λ , to obtain the correct '*photometric*' distance.

$$M_\lambda^0 = m_\lambda - A_\lambda - 5 \log d[\text{pc}] + 5 \quad (1.13)$$

1.3 Interstellar Reddening

In photometric observations, the electromagnetic flux of starlight is quantified by counting the incoming photons within a specific wavelength range of the spectrum. Intervening gas in interstellar space absorbs most of the high-energy photons (X-rays and extreme ultraviolet) especially and scatters much of the light in the B and V bands. Radiation with wavelengths longer than the size of dust particles can pass through the interstellar medium largely unaffected, making the sky nearly transparent to infrared, microwave, and radio waves. The diminishing of short-wavelength luminosity makes the source appear redder than its intrinsic color, giving rise to the effect known as reddening.

This phenomenon can be illustrated by comparing the observed SED of stellar light to the theoretical SED predicted by Planck's Law, as shown in Figure 1.3.

Let B and V be the apparent luminosities of a star as received by the observer, while B_0 and V_0 represent the unextincted luminosities. Then, the effect of extinction can be formulated as follows:

$$A_B = B - B_0 \quad (1.14)$$

$$A_V = V - V_0 \quad (1.15)$$

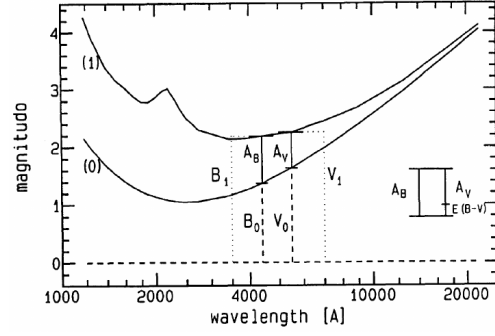


Figure 1.3: Comparison between emitted (0) and observed (1) SEDs. For given band, deviation between the curves represents interstellar extinction (here, A_B and A_V). On the right side, the relative difference between extinction of two bands depicting the definition of interstellar reddening $E(B-V)$. (Krelowski and Papaj, 1993)

From the plot, it can be noticed that B band experienced higher extinction than V, skewing the SED towards red. The difference between the two extinctions termed as *color excess* - that is, the deviation of a star's apparent color from its intrinsic color.

$$E(B - V) = A_B - A_V = (B - V) - (B - V)_0 \quad (1.16)$$

Color excess is more widely used than extinction to describe the effect of the ISM on starlight, as it can be measured photometrically and do not depend on the measurement of distance to the target.

1.3.1 Reddening Ratio

The ratio of total-to-selective absorption, R_λ , summarizes the effect of extinction and reddening on the intensity of light along the line-of-sight. It is formulated as:

$$R_\lambda^{BV} = \frac{A_\lambda}{E(B - V)} \quad (1.17)$$

For Milky Way, the reddening ratio in V-band, R_V , is evaluated about 3.23 (Sandage et al., 2004), whereas in B-band it set to be as $R_B = R_V + 1$

Rearranging the above relation and writting it as:

$$A_\lambda = R_\lambda E_{BV} = R_\lambda (A_B - A_V) \quad (1.18)$$

The above equation is used for transforming measured color excess into extinction for any band, provided R_λ is priorly known. Rearranging it further, for V-band, the equation takes the form:

$$\frac{A_B}{A_V} = 1 + \frac{1}{R_V} \quad (1.19)$$

Cardelli et al. (1989) generalized the extinction law in this form for all the bands with V-band as reference as discussed in the following section.

Another important aspect of the reddening ratio is to convert color excess from one color basis, say E_{BV} to another, E_{12} . It can be done as follows:

$$E_{12} = (R_1^{BV} - R_2^{BV})E_{BV}$$

This leads the transformation law for reddening ratio R_λ :

$$R_\lambda^{12} = \frac{R_\lambda^{BV}}{R_1^{BV} - R_2^{BV}} \quad (1.20)$$

The values of R_λ^{BV} are calculated by Fouqué et al. (2007) and given in Table 1.1.

1.3.2 Interstellar Extinction Law

Observations of the spectral energy distribution (SED) of stellar light by Whiteoak (1966) demonstrated the diminishing effect of interstellar extinction at longer wavelengths. This finding led to the formulation of an extinction law - a relation between extinction ratio and wavelength of stellar light. In its simplest form, the extinction law formulated as :

$$\frac{A(\lambda)}{A(V)} = a(\lambda^{-1}) + \frac{b(\lambda^{-1})}{R_V} \quad (1.21)$$

Here $a(\lambda^{-1})$ and $b(\lambda^{-1})$ are determined empirically for different spectral ranges (Cardelli et al., 1989). Since, historically, observational data in the V-band were available for most

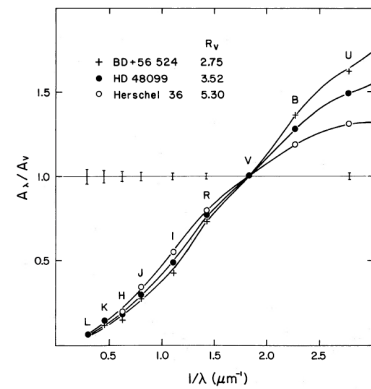


Figure 1.4: Galactic extinction law fitted for UBRIJHKL along three directions in the sky, with varying R_V , with A_V as reference. (Cardelli et al., 1989)

stars, extinction in the V-band has been used as a reference. The Figure 1.4 depicts that a variation in R_V leads to changes in extinction law, where short wavelength have significantly affected as compared to longer wavelengths. Using the averaged value over all the directions in the Milky Way for reddening ratio ($R_V = 3.23$) in the formulation (1.21), [Fouqué et al. \(2007\)](#) evaluated the extinction law for BVRIJHK bands as given in the Table 1.1

Table 1.1: Extinction Law and corresponding reddening ratio adopted from Table 4 of [Fouqué et al. \(2007\)](#)

Band filters	Effective Wavelength λ (μm)	Extinction Law $\frac{A(\lambda)}{A(V)} = a_\lambda + \frac{b_\lambda}{R_V}$	Reddening ratio $R_\lambda = \frac{A_\lambda}{A_V} \times R_V$
B	0.438	1.31	4.231
V	0.545	1	3.230
I	0.798	0.608	1.963
J	1.235	0.292	0.943
H	1.662	0.181	0.584
K	2.159	0.119	0.384

1.3.3 Wesenheit Function: Reddening-free Magnitude

To mitigate the effects of interstellar extinction, [Madore \(1982\)](#) introduced a multiband pseudo-magnitude based on the reddening ratio, termed as the *Wesenheit magnitude*. By construction, this quantity is insensitive to interstellar reddening, provided that the reddening law is known accurately. Its formulation proceeds as follows.

The reddening ratio is defined as

$$\begin{aligned}
 R_\lambda^{12} &= \frac{A_\lambda}{E_{12}} \\
 &= \frac{m_\lambda - m_\lambda^0}{(m_1 - m_2) - (m_1 - m_2)_0},
 \end{aligned} \tag{1.22}$$

where m_λ and m_λ^0 are the apparent and unreddened magnitudes, respectively.

Rearranging the above expression yields

$$m_\lambda - R_\lambda^{12}(m_1 - m_2) = m_\lambda^0 - R_\lambda^{12}(m_1 - m_2)_0. \tag{1.23}$$

All terms on the left-hand side, except for R_λ^{12} , are observable quantities. The equality demonstrates that this linear combination of apparent magnitudes is equivalent to the corresponding intrinsic quantity, implying that it is unaffected by reddening. Consequently, if the reddening law is known with sufficient precision, the Wesenheit magni-

tude attains identical values for reddened and unreddened sources, thereby constituting a reddening-free observable.

Note 1.3.1: Wesenheit Function - Equivalence Relation

For the Wesenheit magnitude, the following relation holds for any color index ($m_1 - m_2$):

$$W_{\lambda}^{12} = m_{\lambda} - R_{\lambda}^{12}(m_1 - m_2) = W_0^{12}. \quad (1.24)$$

This equality is violated if the adopted reddening law, extinction, or color excess is incorrect and can only be satisfied when the errors in these parameters are related as follows:

$$\delta A_{\lambda} - R_{\lambda}^{12} \delta E_{12} - \delta R_{\lambda}^{12} E_{12} = 0. \quad (1.25)$$

A comparison between the definitions of the intrinsic absolute magnitude and the Wesenheit magnitude clarifies their respective dependencies:

$$M_{\lambda}^0 = m_{\lambda} - R_{\lambda}^{BV} E(B - V) - \mu, \quad (1.26)$$

$$W_{\lambda}^{BV} = m_{\lambda} - R_{\lambda}^{BV} (B - V) - \mu, \quad (1.27)$$

where μ denotes the distance modulus.

The absolute magnitude depends on four independent quantities: apparent magnitude, reddening ratio, color excess, and distance modulus. In contrast, the Wesenheit magnitude depends only on the apparent magnitude, reddening ratio, and distance modulus, and is independent of the color excess. Although defined for a specific color index, the Wesenheit formalism naturally forms a family of reddening-free magnitudes. For example, BVIJHK photometry allows fifteen independent color combinations, yielding fifteen Wesenheit families per band and a total of ninety reddening-free magnitudes derived from six apparent magnitudes.

For a given photometric band, consistency among the values obtained from different Wesenheit families is expected if the adopted extinction law is correct. This property provides a means to test and calibrate the reddening law using multi-band photometry, an approach that is applied to the present dataset and discussed further in Chapter 4.

Finally, the propagation of measurement uncertainties in absolute and Wesenheit mag-

nitudes is considered. With multi-epoch photometry, the apparent magnitudes can be determined with negligible uncertainty ($\delta m_\lambda = 0$), and assuming an accurate reddening law ($\delta R_\lambda = 0$), the dominant sources of error are the distance modulus ($\delta\mu$) and the color excess ($\delta E(B - V)$). From Eqs. (1.26) and (1.27), one obtains

$$\delta M_\lambda = - (R_\lambda^{BV} \delta E(B - V) + \delta\mu), \quad (1.28)$$

$$\delta W_\lambda^{BV} = -\delta\mu. \quad (1.29)$$

Both absolute and Wesenheit magnitudes are equally sensitive to uncertainties in the distance modulus. However, the Wesenheit magnitude is immune to errors in the color excess, a property that is central to this work as it enables a clean separation between distance- and reddening-related uncertainties for individual Cepheids.

$$\bar{W}_\lambda^{12} = \alpha_\lambda^{12}(\log P - 1) + \beta_\lambda^{12}$$

$$\bar{M}_\lambda = \alpha_\lambda(\log P - 1) + \beta_\lambda$$

$$\Delta W_\lambda^{12} = W_\lambda^{12} - \bar{W}_\lambda^{12}$$

$$\Delta M_\lambda = M_\lambda^0 - \bar{M}_\lambda$$

The application of these relations to the calibration of the Leavitt Law is presented in *Part II* of this thesis.

1.4 Distance

The vast scale of the Universe necessitates the use of specialized distance measures. Astronomical units are used within the Solar System, light-years or parsecs on stellar and Galactic scales, and parsec multiples for extragalactic distances, with redshift serving as a cosmological distance indicator in an expanding Universe. The relationships among these commonly used units are summarized in Table 1.2.

Table 1.2: Conversion of astronomical distance units

	Kilometer	Astronomical Unit	Light Year	Parsec
Kilometer	1			
Astronomical Unit	1.49×10^8	1		
Light Year	9.46×10^{12}	6.32×10^4	1	
Parsec	3.08×10^{13}	2.06×10^5	3.26	1

Accurate distance measurements are essential in astronomy, as derived physical properties and our understanding of Galactic and cosmic structure depend critically on them.

Because no single method is valid across all distance scales, different techniques are employed depending on the target and spatial regime. This section describes two geometrical distance determination methods—trigonometric parallax, the Infrared Surface Brightness (IRSB) technique and cluster membership method — used to infer distances to Galactic Cepheid variable stars.

1.4.1 Parallax Method

Trigonometric parallax is one of the most fundamental techniques for astronomical distance determination. In a right-angled triangle, knowledge of one side and a non-right angle uniquely determines the remaining sides through trigonometric relations. Since these relations depend only on angular measurements and not on the absolute scale of the triangle, they are particularly well suited to astronomical applications where direct distance measurements are infeasible.

In stellar parallax, a conceptual triangle is defined by the Earth, the Sun, and a nearby star. The baseline of this triangle is the Earth–Sun separation, equal to 1 astronomical unit (AU). As the Earth moves along its orbit, observations taken six months apart reveal an apparent shift in the position of the star relative to distant background sources. This angular displacement, known as the parallax angle, is extremely small and is therefore measured in milliarcseconds (mas).

The distance to a star can then be determined from the measured parallax angle using the tangent trigonometric relation:

$$\tan p = \frac{1AU}{d} \quad (1.30)$$

For very small angles (as in astronomical observations), $\tan p$ is approximately equal to p when measured in radians. Therefore, the distance formula can be simplified to:

$$d = \lim_{p \rightarrow 0} \frac{1AU}{\tan p} = \frac{1AU}{p} \quad (1.31)$$

To standardize distance measurements, astronomers use the parsec as a unit of length.

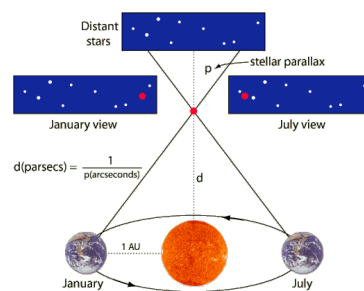


Figure 1.5: As Earth orbits the Sun, nearby stars appear to shift against distant background stars—this shift, called parallax, is used to measure their distance using trigonometric ratio.

Source: hyperphysics.phy-astr.gsu.edu

1 parsec (pc) is defined as the distance at which 1 AU subtends an angle of 1 arcsecond (1/3600 of a degree).

In this thesis, parallax-based distances from *Gaia* are adopted; hence, it is important to consider systematic uncertainties. Being bright yellow giants, Cepheid variables saturate CCD pixels, leading to uncertain parallax measurements [Lindgren \(2018\)](#). The marginal impact of these parallax uncertainties on the resulting Leavitt Law is discussed in a later chapter.

1.4.2 Infrared Surface Brightness Method

The Infrared Surface Brightness (IRSB) method is a geometrical technique belonging to the broader class of *parallax of pulsation* methods. Its fundamental principle is to determine the distance to a Cepheid by comparing its *linear radius variation*, inferred from radial velocity measurements, with its *angular diameter variation*, derived from photometry.

For a spherical star at distance d , the basic relation is

$$d = \frac{\text{linear diameter}}{\text{angular diameter}}. \quad (1.32)$$

The IRSB method has its roots in the classical Baade–Becker–Wesselink (BBW) technique [Baade \(1926\)](#); [Becker \(1940\)](#); [Wesselink \(1946\)](#), which exploits the existence of a linear relation between surface brightness and stellar color. [Barnes and Evans \(1976\)](#) were the first to apply this method to Cepheids using optical photometry (V and $V-R$) for determining their distances.

In the Barnes–Evans formulation, the visual surface brightness parameter F_V is defined such that

$$V_0 = F_V - 5 \log R + C, \quad (1.33)$$

where V_0 is the dereddened apparent magnitude, R is the stellar radius (in solar units), and C is a constant that depends on the choice of units.

The surface brightness parameter is empirically found to depend linearly on colour,

$$F_V = a(V - R)_0 + b, \quad (1.34)$$

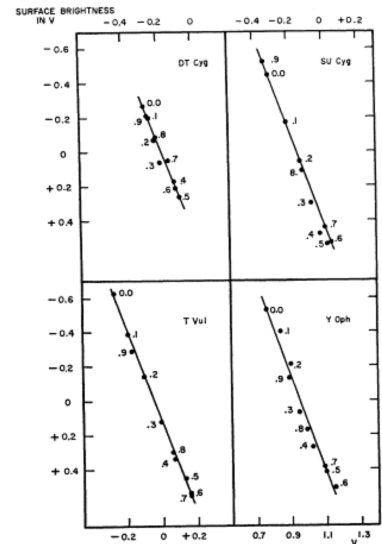


FIG. 5. Surface brightness in V as a function of the color V.

Figure 1.6: Linear relation between surface brightness and color. [Svolopoulos \(1960\)](#)

where $(V - R)_0$ is the dereddened colour index.

Let the stellar radius vary with pulsation phase ϕ as

$$R(\phi) = R_0 + \Delta R(\phi), \quad (1.35)$$

where R_0 is the mean radius. At two pulsation phases ϕ_1 and ϕ_2 with identical color $(V - K)_0$, the surface brightness parameter F_V is the same. Therefore,

$$V_1 = F_V - 5 \log(R_0 + \Delta R_1) + C, \quad (1.36)$$

$$V_2 = F_V - 5 \log(R_0 + \Delta R_2) + C, \quad (1.37)$$

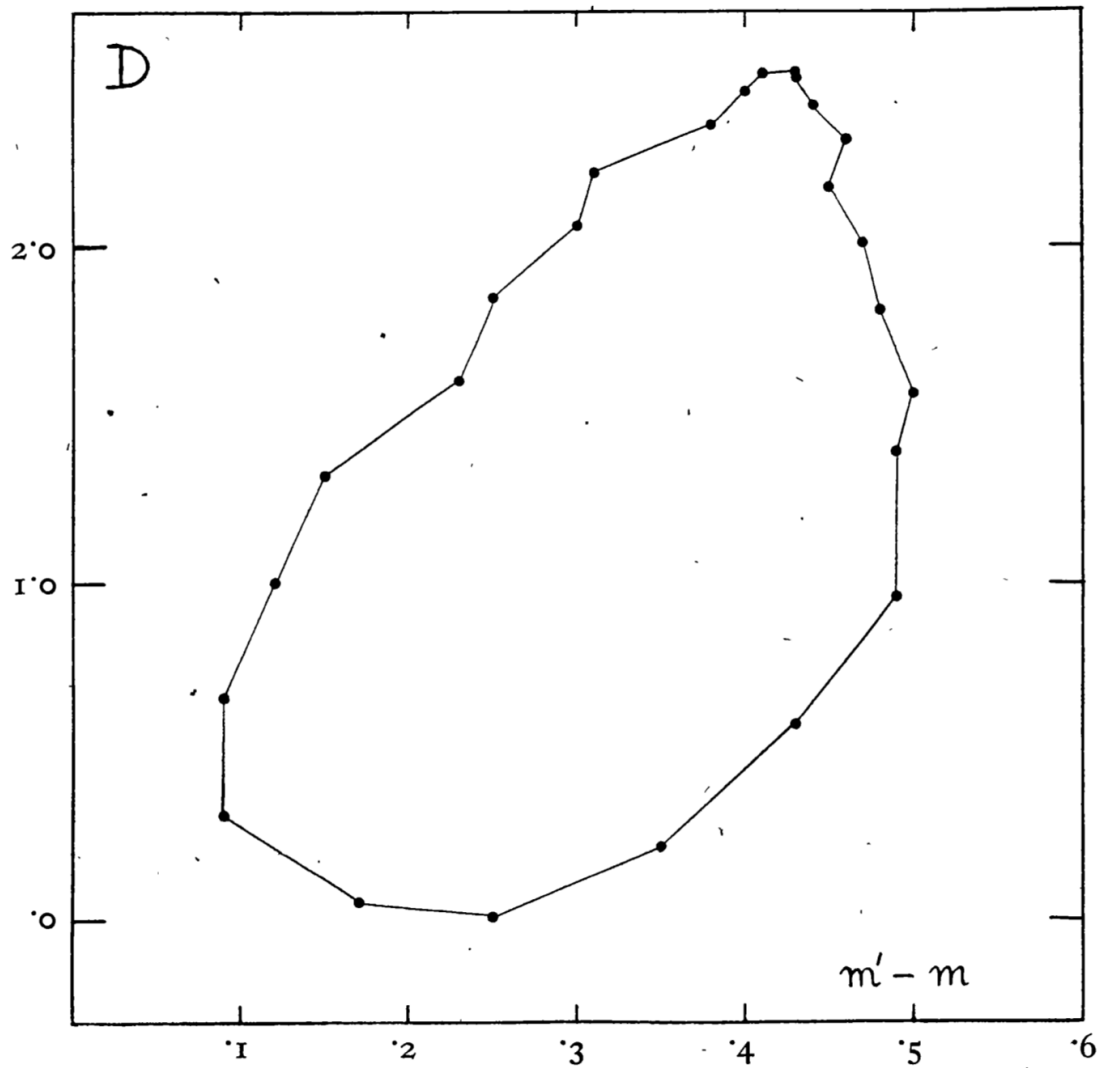


Figure 1.7: Figure 3 of [Wesselink \(1946\)](#)

Subtracting the two equations gives

$$V_2 - V_1 = 5 \log \left(\frac{R_0 + \Delta R_1}{R_0 + \Delta R_2} \right). \quad (1.38)$$

Defining $H = 10^{0.2(V_2 - V_1)}$, the mean radius can be written as

$$R_0 = \frac{H \Delta R_2 - \Delta R_1}{1 - H}. \quad (1.39)$$

The linear radius variation $\Delta R(\phi)$ is obtained independently from spectroscopic radial velocity measurements. The observed radial velocity $v_r(t)$ is converted into the pulsational velocity of the photosphere $v_p(t)$ using the projection factor p ,

$$v_p(t) = p[v_r(t) - v_\gamma], \quad (1.40)$$

where v_γ is the systemic velocity. The projection factor accounts for geometric projection effects, limb darkening, atmospheric velocity gradients, and phase-dependent photospheric motion.

The radial displacement is obtained by integrating the pulsation velocity,

$$\Delta R = \int_{t_0}^t v_p(t') dt'. \quad (1.41)$$

Since Cepheid pulsation is periodic with period P , it is convenient to express the integration in terms of pulsation phase,

$$\phi = \frac{t - t_0}{P}, \quad dt = P d\phi, \quad (1.42)$$

leading to

$$\Delta R(\phi) = pP \int_0^\phi [v_r(\phi') - v_\gamma] d\phi'. \quad (1.43)$$

The angular diameter can also be written geometrically as

$$\theta = \frac{2(R_0 + \Delta R(\phi))}{d} \quad (1.44)$$

$$\theta(\phi) = 10^{4.2207 - 0.1 V_0(\phi) - F_V(\phi)}$$

$$\theta(\phi) = \frac{2R_0}{d} + \frac{2\Delta R(\phi)}{d} \quad (1.45)$$

$$\theta(\phi) = \frac{2R_0}{d} + \frac{2pP}{d} \int_0^\phi [v_r(\phi') - v_\gamma] d\phi'. \quad (1.46)$$

where θ is expressed in milliarcseconds.

This expression is linear and may be written as

$$\theta(\phi) = A + B X(\phi), \quad (1.47)$$

where both $\theta(\phi)$ and $X(\phi)$ are directly derived from observations. The slope B yields the distance,

$$d = \frac{2pP}{B}. \quad (1.48)$$

Subsequent improvements were achieved by adopting the $(V - K)$ colour [Welch \(1994\)](#), which provides a larger temperature leverage: the V band lies near the Wien regime of Cepheid spectral energy distributions, while the K band lies on the Rayleigh–Jeans tail, making the color primarily sensitive to effective temperature. In addition, extinction effects are significantly reduced in the K band.

A tight empirical calibration between $(V - K)$ and the surface brightness parameter F_V , based on interferometric angular diameter measurements, was established by [Fouque and Gieren \(1997\)](#). This calibration forms the basis of the modern Infrared Surface Brightness (IRSB) method, which was further refined and applied to Cepheids by [Storm et al. \(2011\)](#).

1.4.3 Cluster Membership Method

Consider an open star cluster containing a Classical Cepheid variable. All bona fide cluster members share a common true parallax π_{cl} . Let N cluster stars have measured parallaxes $\{\pi_i\}_{i=1}^N$.

Each parallax measurement is modeled as

$$\pi_i = \pi_{\text{cl}} + \epsilon_i, \quad (1.49)$$

where the errors ϵ_i satisfy

$$\mathbb{E}[\epsilon_i] = 0, \quad \text{Var}(\epsilon_i) = \sigma_i^2, \quad (1.50)$$

and are assumed to be statistically independent and Gaussian.

The goal is to estimate π_{cl} with minimum uncertainty and assign this distance to the Cepheid.

Likelihood Function

The joint likelihood of observing the parallaxes $\{\pi_i\}$ given π_{cl} is

$$\mathcal{L}(\pi_{\text{cl}}) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left[-\frac{(\pi_i - \pi_{\text{cl}})^2}{2\sigma_i^2}\right]. \quad (1.51)$$

Taking the logarithm,

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i=1}^N \left[\frac{(\pi_i - \pi_{\text{cl}})^2}{\sigma_i^2} + \ln(2\pi\sigma_i^2) \right]. \quad (1.52)$$

The logarithmic normalization term does not depend on π_{cl} and may be ignored for maximization.

Maximum Likelihood Estimator

Differentiating the log-likelihood with respect to π_{cl} and setting the derivative to zero yields

$$\frac{\partial \ln \mathcal{L}}{\partial \pi_{\text{cl}}} = \sum_{i=1}^N \frac{\pi_i - \pi_{\text{cl}}}{\sigma_i^2} = 0. \quad (1.53)$$

Rearranging terms,

$$\sum_{i=1}^N \frac{\pi_i}{\sigma_i^2} = \pi_{\text{cl}} \sum_{i=1}^N \frac{1}{\sigma_i^2}. \quad (1.54)$$

Solving for π_{cl} , the maximum-likelihood estimator is

$$\hat{\pi}_{\text{cl}} = \frac{\sum_{i=1}^N \pi_i / \sigma_i^2}{\sum_{i=1}^N 1 / \sigma_i^2} \quad (1.55)$$

which is the weighted mean parallax.

The expectation value of the estimator is

$$\mathbb{E}[\hat{\pi}_{\text{cl}}] = \frac{\sum_{i=1}^N \mathbb{E}[\pi_i] / \sigma_i^2}{\sum_{i=1}^N 1 / \sigma_i^2} \quad (1.56)$$

$$= \frac{\sum_{i=1}^N \pi_{\text{cl}} / \sigma_i^2}{\sum_{i=1}^N 1 / \sigma_i^2} = \pi_{\text{cl}}. \quad (1.57)$$

Thus, the estimator is unbiased.

Variance of the Estimator

Define weights $w_i = 1/\sigma_i^2$. The estimator can be written as

$$\hat{\pi}_{\text{cl}} = \frac{\sum_i w_i \pi_i}{\sum_i w_i}. \quad (1.58)$$

The variance is

$$\text{Var}(\hat{\pi}_{\text{cl}}) = \frac{\sum_i w_i^2 \sigma_i^2}{(\sum_i w_i)^2} \quad (1.59)$$

$$= \frac{\sum_i (1/\sigma_i^2)}{(\sum_i 1/\sigma_i^2)^2} \quad (1.60)$$

$$= \left[\left(\sum_{i=1}^N \frac{1}{\sigma_i^2} \right)^{-1} \right]. \quad (1.61)$$

Hence, the standard deviation is

$$\sigma_{\hat{\pi}} = \left(\sum_{i=1}^N \frac{1}{\sigma_i^2} \right)^{-1/2}. \quad (1.62)$$

By the Gauss–Markov theorem, among all linear unbiased estimators of π_{cl} , the weighted mean with weights $w_i = 1/\sigma_i^2$ has the minimum variance.

Therefore, $\hat{\pi}_{\text{cl}}$ is the optimal estimator.

The cluster distance is obtained from the parallax as

$$d = \frac{1}{\hat{\pi}_{\text{cl}}}, \quad (1.63)$$

with fractional uncertainty

$$\frac{\sigma_d}{d} \approx \frac{\sigma_{\hat{\pi}}}{\hat{\pi}_{\text{cl}}}. \quad (1.64)$$

Since the Cepheid is a cluster member, this distance applies directly to the Cepheid and can be used to calibrate the period–luminosity relation.

The cluster Cepheid parallax method follows rigorously from maximum likelihood estimation under Gaussian errors. The weighted mean parallax is unbiased, minimum-variance, and provides a high-precision geometric distance to Cepheid variables.

1.5 Distance Modulus (μ)

For the ease of calculation, a dimensionless measure of distance is used in astronomy called as the distance modulus, μ , which provides a logarithmic transformation of the

distance to an object, by setting 10 parsecs as the reference distance. It can also be expressed as the difference between the apparent magnitude and the absolute magnitude of an object as the absolute magnitude is the measure of luminosity at the 10 pc distance from the source. Mathematically, distance modulus can be formulated as:

$$\mu = m - M = 5 \log_{10} \left[\frac{d}{10pc} \right]$$

Part II

Calibration Method and Dataset

2 Calibration Method

2.1 Leavitt Law and their Wesenheits

Observations in different bands of light provide information about the effect of interstellar reddening on light. For longer wavelengths, the impact of reddening decreases significantly, implying a reduced possibility of reddening errors and leading to a tighter Leavitt Law. In its most generic form, the linear fit to the dataset is expressed as follows:

$$M_\lambda = \alpha_\lambda(\log P - 1) + \beta_\lambda \quad (2.1)$$

$$W_\lambda^{12} = \alpha_\lambda^{12}(\log P - 1) + \beta_\lambda^{12} \quad (2.2)$$

Since the periodic range of Cepheids lies between 0.49 and 1.8 on a logarithmic scale, 1 is chosen as the pivot point and subtracted from the period before calculating the regression coefficients, i.e., the slope and intercept. The Leavitt Law for each band is shown in Figure 2.8, including four selected Wesenheit variants for VI, VK, IH, and JK.

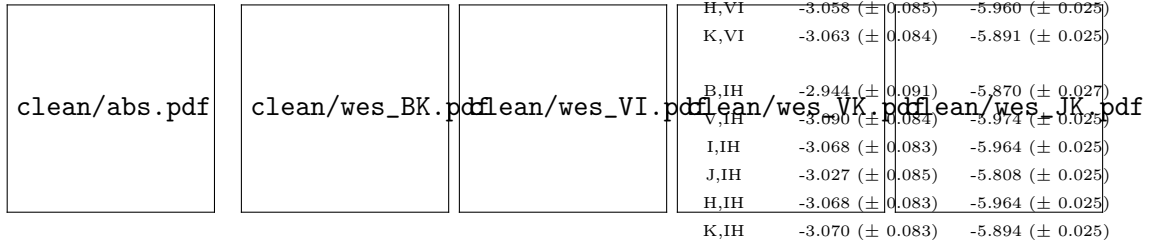


Figure 2.1: BVIJHK PL and PW relations with their Pearson correlation coefficient (r-value).

The plots above provide a comparative overview

Table 2.1: PL relations are shown in six bands, along with the corresponding PW relations in four colors. Notice the decreasing error in the slope and intercept with increasing wavelength.

Band	slope (error)	intercept (error)
M_λ	α_λ	β_λ
B	-1.855 ($\pm$ 0.110)	-3.221 ($\pm$ 0.033)
V	-2.259 ($\pm$ 0.098)	-3.951 ($\pm$ 0.029)
I	-2.563 ($\pm$ 0.093)	-4.735 ($\pm$ 0.028)
J	-2.784 ($\pm$ 0.088)	-5.218 ($\pm$ 0.026)
H	-2.917 ($\pm$ 0.084)	-5.598 ($\pm$ 0.025)
K	-2.971 ($\pm$ 0.084)	-5.653 ($\pm$ 0.025)
Wesenheit	slope (error)	intercept (error)
W_λ	α_λ^{12}	β_λ^{12}
B,BI	-3.175 ($\pm$ 0.095)	-6.046 ($\pm$ 0.028)
V,BI	-3.267 ($\pm$ 0.095)	-6.108 ($\pm$ 0.028)
I,BI	-3.175 ($\pm$ 0.095)	-6.046 ($\pm$ 0.028)
J,BI	-3.078 ($\pm$ 0.088)	-5.847 ($\pm$ 0.026)
H,BI	-3.100 ($\pm$ 0.085)	-5.988 ($\pm$ 0.025)
K,BI	-3.091 ($\pm$ 0.084)	-5.910 ($\pm$ 0.025)
B,VI	-2.869 ($\pm$ 0.101)	-5.838 ($\pm$ 0.030)
V,VI	-3.033 ($\pm$ 0.096)	-5.949 ($\pm$ 0.029)
I,VI	-3.033 ($\pm$ 0.096)	-5.949 ($\pm$ 0.029)
J,VI	-3.010 ($\pm$ 0.088)	-5.801 ($\pm$ 0.026)
H,VI	-3.058 ($\pm$ 0.085)	-5.960 ($\pm$ 0.025)
K,VI	-3.063 ($\pm$ 0.084)	-5.891 ($\pm$ 0.025)
B,IH	-2.944 ($\pm$ 0.091)	-5.870 ($\pm$ 0.027)
V,IH	-3.090 ($\pm$ 0.084)	-5.974 ($\pm$ 0.025)
I,IH	-3.068 ($\pm$ 0.083)	-5.964 ($\pm$ 0.025)
J,IH	-3.027 ($\pm$ 0.085)	-5.808 ($\pm$ 0.025)
H,IH	-3.068 ($\pm$ 0.083)	-5.964 ($\pm$ 0.025)
K,IH	-3.070 ($\pm$ 0.083)	-5.894 ($\pm$ 0.025)
B,JK	-3.269 ($\pm$ 0.089)	-6.517 ($\pm$ 0.027)
V,JK	-3.338 ($\pm$ 0.082)	-6.468 ($\pm$ 0.024)
I,JK	-3.219 ($\pm$ 0.081)	-6.265 ($\pm$ 0.024)
J,JK	-3.099 ($\pm$ 0.082)	-5.952 ($\pm$ 0.024)
H,JK	-3.113 ($\pm$ 0.081)	-6.054 ($\pm$ 0.024)
K,JK	-3.099 ($\pm$ 0.082)	-5.952 ($\pm$ 0.024)

of the Leavitt Laws across different bands. In the first column, notice the decreasing width of the PL relation from the B-band to the K-band. The increasing Pearson correlation coefficient indicates a reduction in residuals. Assuming Cepheids align perfectly with the PL relation suggests that the residuals must stem from errors in reddening or distance measurements.

The contribution of residuals due to extinction errors varies with the band, while errors from the distance modulus remain constant. Therefore, the decrease in the width of the PL relation is strongly related to incorrect reddening measurements.

The four variants of the PW relation have relatively higher r-values (approaching -1) than their

corresponding PL relations, as Wesenheits are independent of extinction-based errors. Being reddening-free, the scatter in the Wesenheit-based Leavitt Law primarily arises from errors in distance and the reddening ratio. In this research, the reddening ratio is assumed to be correctly known ($R_V = 3.23$, [Sandage et al. \(2004\)](#)), making distance the only source of the scatter.

The slope and intercept of each PL and PW relations are summarized in Table 2.1 and Figure 2.2. The Table 2.1 also list the uncertainty in slope and intercept of PL and PW relation, arising due to a larger scatter. After the calibration, these uncertainty will be deminished significantly.

On the other hand, note the convergence of slope and intercept of PW relations at K band for VI, VK and IH in the adjacent figure. JK based PW relation showing a fine deviation from the rest. These subtle variation in PW slope and intercept suggests a fine-tuning of reddening law for longer wavelength, though such exercise is out of the scope of this thesis. In the next section, procedure for analysing these residual of PL and PW relations is briefly discussed.

clean/PLW_mc.pdf

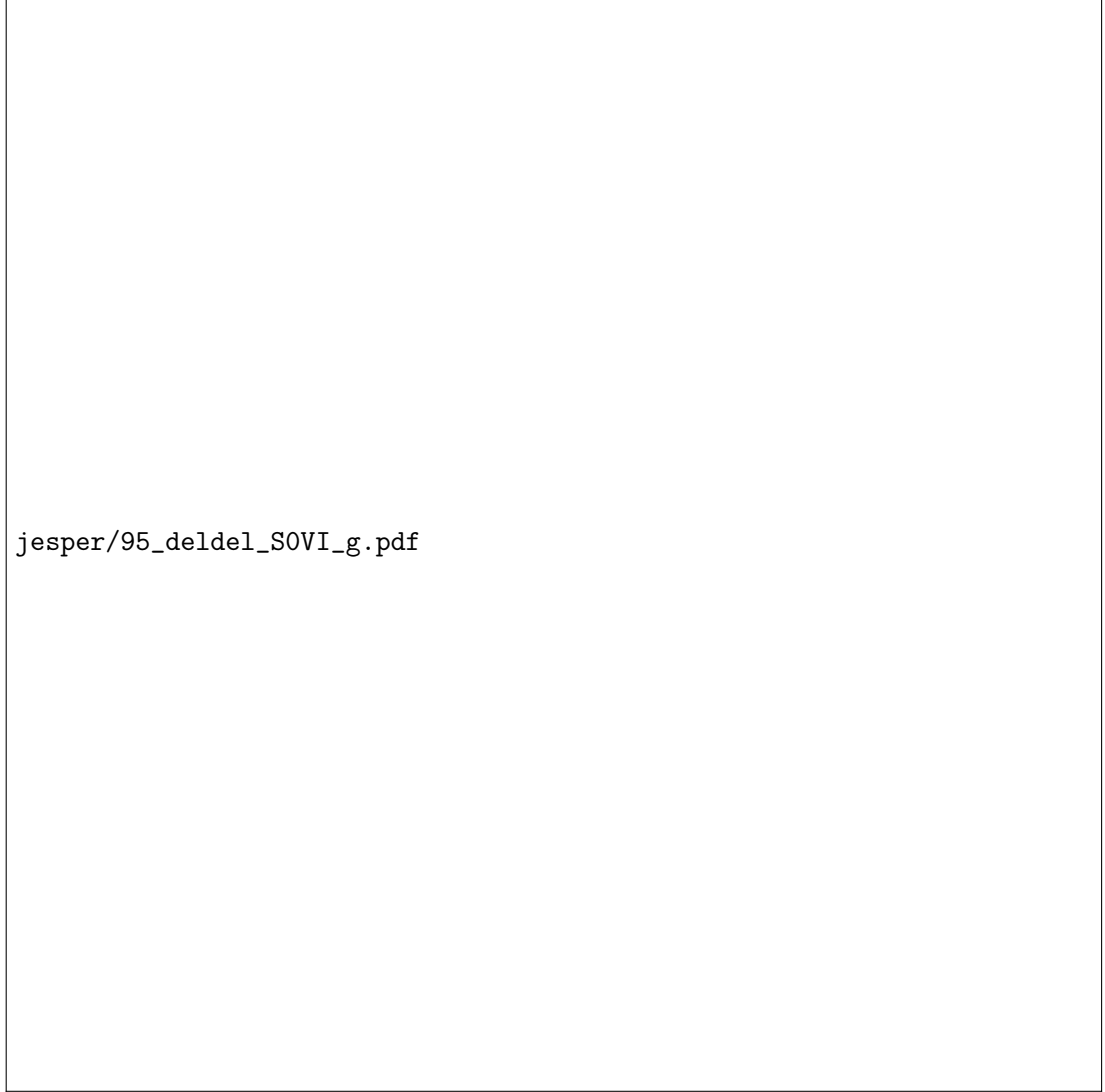
Figure 2.2: Slope and Intercept of BVIJHK PL and PW relations. Interestingly, PW seems constant for all the bands, which is not the case for PL relations.

2.2 Residual Analysis

The residuals of PL and PW relations are calculated as follows:

$$\Delta M_\lambda = M_\lambda - (\alpha_\lambda \log P + \beta_\lambda) \quad (2.3)$$

$$\Delta W_\lambda^{12} = W_\lambda^{12} - (\alpha_\lambda^{12} \log P + \beta_\lambda^{12}) \quad (2.4)$$



(a) ΔW_λ^{VI} - ΔM_λ correlation

(b) ΔW_λ^{JK} - ΔM_λ correlation

Figure 2.3: ΔW - ΔM correlation plots shown for three versions of wesensheit - VI and JK. With increasing wavelength, slope is approaching to 1.

Here, ΔM_λ represents the residuals of the PL relation, and ΔW_λ^{12} represents the residuals of the PW relation. Since distance modulus μ term is involved in both the parameters, M_λ and W_λ , thereby $\delta\mu$, impacts both ΔW and ΔM equally. However, δE_{BV} does not contribute to ΔW , and only affects ΔM by an amount of $\delta E_{BV}/R_\lambda = \delta A_\lambda$. Since other possible factor like the effect of metallicity is not considered in this work, the equation for PL and PW residuals, equations 1.28 and 1.29, can be expressed as:

$$\Delta M_\lambda = -(R_\lambda^{BV} * \delta E(B - V) + \delta\mu)$$

$$\Delta W_\lambda^{BV} = -\delta\mu$$

Geometrically, these equations suggest that δE_{BV} (scaled by R_λ^{-1}) shifts a star along the y-axis (vertically), while $\delta\mu$ affects both the x and y axes equally, moving the star along a line with slope 1. For the complete ensemble of stars, the slope of the residual correlation, ρ , when deviating from 1, suggests a significant contribution from reddening errors, as clearly observed for the case of the B and V bands in Figure 2.3. For the longer wavelengths—J, H, and K—the contribution from reddening errors is minimized, and distance-based errors become dominant, causing the slope ρ to approach 1.

Tracing an individual Cepheid in every ΔW_λ^{12} - ΔM_λ correlation plots provides valuable insight into the extinction error associated with each band. By converting the extinction error into reddening error for each band and averaging these val-

ues, an estimate of the reddening correction for that specific star is obtained. However, at this stage, the correction for the modulus is not yet determined. To estimate the modulus error, a series of trials for $\delta\mu$ are conducted, which shifts the star along the line of slope 1, as shown in Figure 2.4 for stars B and C. The solid black diagonal line represents the regression line derived from one of the $\Delta W - \Delta M$ plots. The vertical deviation from this regression line, for each trial, reflects the amount of reddening correction needed. Several trial combinations for star B are labeled as 'B0' and 'B1', while 'C2', 'C3', and 'C4' correspond to the trial positions for star C.

Figure 2.4: For each star, multiple trials are conducted to estimate the reddening-distance error pair in the B band. This process is then repeated for the remaining bands to constrain the correct $\delta\mu - \delta E_{BV}$ error pair. [Madore et al. \(2017\)](#)

These examples illustrate multiple possible error pairs for the B-band. Similar correlation plots are also constructed for the VIJHK bands. Each residual correlation plot provides the corresponding reddening errors for the respective $\delta\mu$ trials, i.e. every $\delta\mu$ trail corresponds to six δE_{BV} . The $\delta\mu$ trial that minimizes the variance of δE_{BV} across the bands identifies the correct distance correction trial. This ultimately decouples the error budget between $\delta\mu$ and δE_{BV} for each star.

This method of decoupling errors, introduced by [Madore et al. \(2017\)](#), involves correlating ΔW_V^{VI} with ΔM_λ . In contrast, I have used the composite Wesenheit function, ΔW_λ^{VI} , for each correlation instance. The key motivation behind this choice is the definition of the Wesenheit function, which provides a reddening-free magnitude for the

respective band. Specifically, for the VI color, the reddening-free magnitude for the J band should be W_J^{VI} , not W_V^{VI} . This implies that ΔM_J must be correlated with ΔW_J^{VI} . The slopes of the regression lines for the four families of Wesenheit functions—based on BI, VI, IH, and JK—are summarized in Table 2.2.

Band	ρ_λ^{BI} (error)	ρ_λ^{VI} (error)	ρ_λ^{IH} (error)	ρ_λ^{JK} (error)
B	0.797 (± 0.088)	0.819 (± 0.075)	0.791 (± 0.094)	0.619 (± 0.111)
V	0.818 (± 0.066)	0.847 (± 0.060)	0.882 (± 0.079)	0.778 (± 0.095)
I	0.906 (± 0.041)	0.907 (± 0.036)	1.022 (± 0.049)	0.993 (± 0.060)
J	0.979 (± 0.022)	0.976 (± 0.019)	1.013 (± 0.023)	1.037 (± 0.028)
H	0.986 (± 0.014)	0.984 (± 0.013)	1.007 (± 0.015)	1.023 (± 0.018)
K	0.994 (± 0.009)	0.992 (± 0.008)	1.007 (± 0.010)	1.015 (± 0.012)

Table 2.2: Slopes of residual correlation $\Delta W_\lambda^{12} - \Delta M_\lambda$ for BI, VI, IH and JK. Note the systematic decrease in errors as the slope ρ approaching to 1 with increasing wavelength.

2.3 Decoupling $\delta\mu - \delta E_{BV}$

The core process of the calibration is encapsulated in this step. Up until this point, from the Galactic Cepheid dataset, I have derived multiband Leavitt laws and their corresponding Wesenheit Leavitt laws. The correlation of their residuals leads to the results summarized in Table 2.2, which will be used for decoupling the errors δE_{BV} and $\delta\mu$ for each star in the dataset. Given the high sensitivity of reddening at shorter wavelengths, to estimate a realistic correction for reddening error (δE_{BV}), averaging result from the first four bands — B, V, I, and J would be enough.

Since the vertical deviation from the regression line in $\Delta W_\lambda^{12} - \Delta M_\lambda$ suggested the extinction error, mathematically, it can be formulated as:

$$\delta A_\lambda^0 = \Delta M_\lambda - \rho_\lambda^{12} \Delta W_\lambda^{12} - \sigma_\lambda^{12} \quad (2.5)$$

The superscript '0' on A indicates that the distance error trial is not considered yet. ρ_λ^{12} represents the slope of the ΔW_λ^{12} vs. ΔM_λ plot. Since residuals are dispersed around the origin, intercept of the regression lines σ_λ approach to zero and can be neglected.

To convert the extinction error into reddening error, dividing by R_λ^{BV}

$$\delta E^0(B - V)_\lambda = \frac{\delta A_\lambda^0}{R_\lambda^{BV}} \quad (2.6)$$

By making trials for the error in the modulus, $\delta\mu^i$, the corresponding reddening error, δA_λ^i , is calculated as follows:

$$\begin{aligned} \delta A_\lambda(\delta\mu^i) &= (\Delta M_\lambda + \delta\mu^i) - \rho_\lambda^{12}(\Delta W_\lambda^{12} + \delta\mu^i) \\ &= (\Delta M_\lambda - \rho_\lambda^{12}\Delta W_\lambda^{12}) + \delta\mu^i - \rho_\lambda^{12}\delta\mu^i \\ &= \delta A_\lambda^0 + \delta\mu^i - \rho_\lambda^{12}\delta\mu^i \end{aligned}$$

Therefore, for given modulus correction trail, extinction in any band will be calculated as follow:

$$\delta A_\lambda^i = \delta A_\lambda^0 + \delta\mu^i(1 - \rho_\lambda^{12}) \quad (2.7)$$

$$\delta E_\lambda^i(B - V) = \frac{\delta A_\lambda^i}{R_\lambda^{BV}} \quad (2.8)$$

$$(2.9)$$

Considering ideal case where extinction law $\frac{A_\lambda}{A_V}$ and reddening ratio R_λ are precisely known for the line-of-sight of each Cepheid of the dataset. Let i number of trails made for modulus correction $\delta\mu$ which correspond to six variants of reddening corrections (BVIJHK) for each weseneit-color (BI, VI, IH and JK). Each Cepheid yields an error matrix of 100 rows ($\delta\mu^i$) and 6 x 4 columns for expected $\delta E_\lambda^i(B - V)$.

Let, k^{th} trail be the correct one, i.e. reddening corrections δE_λ suggested by each band must be agreeing with remaining ones., $\delta\mu^k$ must yield the same reddening correction δE^k for all the bands, implying their variance is zero.

$$\delta E_{BV}(\delta\mu^k) = \delta E_B^k = \delta E_V^k = \delta E_I^k = \delta E_J^k = \delta E_H^k = \delta E_K^k = \langle \delta E_\lambda^k \rangle_\lambda \quad (2.10)$$

For dataset, it the variance across the bands being the minimum for the best solution, when .

$$k = \min(\text{var}(\delta E_{\lambda}^i(B - V))_{\lambda})_i \quad (2.11)$$

2.4 Calibrated Leavitt Law

These corrections will be adjusted with the original data as follow.

$$M_{\lambda}^* = M_{\lambda}^0 + \delta A_{\lambda}^* + \delta \mu^*$$

2.5 cluster Cepheid

In the dataset, 17 Cepheids are identified as members of clusters with statistical parallaxes measured by cruz2023. A comparison between the calibrated distances of these Cepheids shows a good agreement in reddening value with $\sigma = 0.049$.

For distance modulus, nine of 17 cluster Cepheid deviated by 0.1 mag only, but five Cepheids (EV Sct, QZ Nor, SV Vul, Y Sct and X Lac) have deviation larger than 0.4 mag.

Gaia parallax based distances are independent of reddening error as the method is fundamentally geometrical in nature. Although, Cepheid being one of the brightest target in the sky, leading to saturation of photons on the CCD causing a strong bias for longer period Cepheids. This makes parallax of Cepheid slightly uncertain as compare to dimmer stars. I have implemented the calibration algorithm on the same set of stars and determine calibrated distance for individual Cepheid. If the algorithm works perfectly, calibrated distance must agree with IRSB based calibrated distance.

9_plots/6_rms/95_0_star_SBI_g.pdf

9_plots/6_rms/95_1_star_SBI_g.pdf

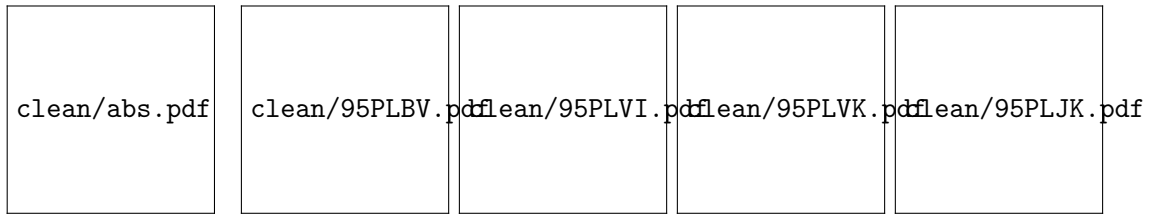
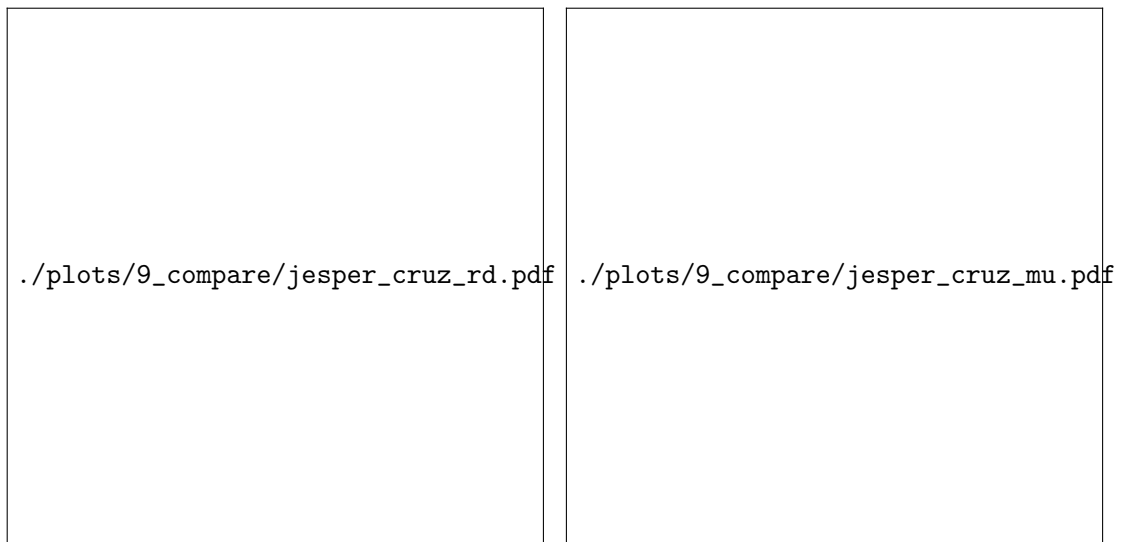


Figure 2.6: BVIJHK PL and PW relations with their Pearson correlation coefficient (r-value).



(a) Reddening correction comparision

(b) Distance modulus comparision

Figure 2.7: Deviation of distance and reddening with 17 Cluster Cepheids (Cruz, 2023)

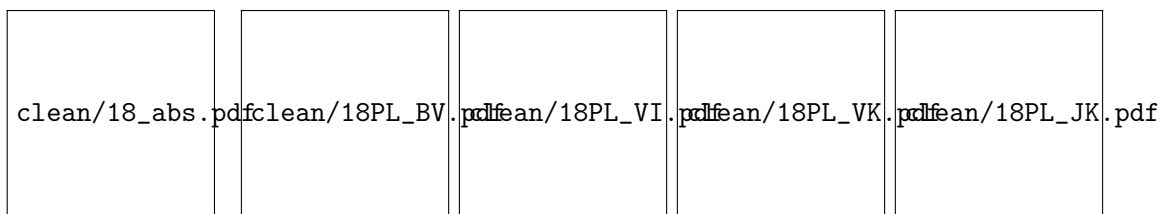


Figure 2.8: BVIJHK PL and PW relations with their Pearson correlation coefficient (r-value).

Part III

Result Implications

Bibliography

- Agency, E. S., editor (1997). *The Hipparcos and Tycho Catalogues*. ESA Publications Division, Noordwijk, The Netherlands. Hipparcos Catalogue (ESA SP-1200).
- Akrami, Y., Ashdown, M., and Aumont, J. (2020). Planck2018 results: Vii. isotropy and statistics of the cmb. *Astronomy & Astrophysics*, 641:A7.
- Alpher, R. A. and Herman, R. (1948). Evolution of the universe. *Nature*, 162(4124):774–775.
- Baade, W. (1926). Über eine Möglichkeit, die Pulsationstheorie der δ Cephei-Veränderlichen zu prüfen. *Astronomische Nachrichten*, 228(20):359.
- Baade, W. (1958). Problems in the determination of the distance of galaxies. *The Astronomical Journal*, 63:207–210.
- Bailer-Jones, C., Rybizki, J., Foesneau, M., Demleitner, M., and Andrae, R. (2021). Estimating distances from parallaxes. v. geometric and photogeometric distances to 1.47 billion stars in gaia early data release 3. *The Astronomical Journal*, 161(3):147.
- Barnes, T. G. and Evans, D. S. (1976). Stellar angular diameters and visual surface brightness - I. Late spectral types. , 174:489–502.
- Becker, W. (1940). Spektralphotometrische Untersuchungen an δ Cephei-Sternen. X. Ein Beitrag zur Prüfung der Pulsationstheorie der δ Cephei-Veränderlichen durch die Beobachtung und eine unabhängige Ableitung der Perioden-Helligkeitsbeziehung. (Mitteilungen des Astrophysikalischen Observatoriums Potsdam, Nr. 4.) Mit 5 Abbildungen. , 19:289.
- Bethe, H. A. (1939). Energy production in stars. *Physical Review*, 55(5):434.
- Cardelli, J. A., Clayton, G. C., and Mathis, J. S. (1989). The relationship between infrared, optical, and ultraviolet extinction. *The Astrophysical Journal*, 345:245–256.

- Christian, D. (2011). *Maps of Time: An Introduction to Big History*. University of California Press, Berkeley, 2 edition. pp. 17–18.
- Clementini, G., Ripepi, V., Garofalo, A., Molinaro, R., Muraveva, T., Leccia, S., Rimoldini, L., Holl, B., de Fombelle, G. J., Sartoretti, P., et al. (2023). Gaia data release 3-specific processing and validation of all-sky rr lyrae and cepheid stars: The rr lyrae sample. *Astronomy & Astrophysics*, 674:A18.
- Colvin, A. Z. (2018). 05-milky way galaxy (lofe05240).png. Wikimedia Commons. Map showing the Milky Way Galaxy in the Orion Arm, licensed under CC-BY-SA 4.0.
- Eddington, A. (1920). The internal constitution of the stars. *The Scientific Monthly*, 11(4):297–303.
- Eddington, A. S. (1917). The pulsation theory of Cepheid variables. *The Observatory*, 40:290–293.
- Einstein, A. (1905). Über einen die erzeugung und verwandlung des liches betreffenden heuristischen gesichtspunkt. *Annalen der Physik*, 17:132–148. English translation: "On a Heuristic Point of View Concerning the Production and Transformation of Light".
- Einstein, A. (1917). Kosmologische betrachtungen zur allgemeinen relativitätstheorie. *Sitzungsberichte der Koniglich Preussischen Akademie der Wissenschaften*, pages 142–152.
- Feast, M. W. and Catchpole, R. M. (1997). The cepheid period–luminosity zero-point from *hipparcos* trigonometrical parallaxes. *Monthly Notices of the Royal Astronomical Society*, 286:L1–L5.
- Fernie, J. (1994). Color excesses for classical cepheids from bv photometry. *The Astrophysical Journal*, vol. 429, no. 2, pt. 1, p. 844–845, 429:844–845.
- Fouqué, P., Arriagada, P., Storm, J., Barnes, T., Nardetto, N., Mérand, A., Kervella, P., Gieren, W., Bersier, D., Benedict, G., et al. (2007). A new calibration of galactic cepheid period-luminosity relations from b to k bands, and a comparison to lmc relations. *Astronomy & Astrophysics*, 476(1):73–81.
- Fouque, P. and Gieren, W. P. (1997). An improved calibration of Cepheid visual and infrared surface brightness relations from accurate angular diameter measurements of cool giants and supergiants. , 320:799–810.

- Fox, O. D., Waczynski, A., Wen, Y., Foltz, R. D., Hill, R. J., Kimble, R. A., Malumuth, E., and Rauscher, B. J. (2009). The ^{55}Fe x-ray energy response of mercury cadmium telluride near-infrared detector arrays. *arXiv e-prints*.
- Freedman, W. L., Madore, B. F., Gibson, B. K., Ferrarese, L., Kelson, D. D., Sakai, S., Mould, J. R., Kennicutt Jr, R. C., Ford, H. C., Graham, J. A., et al. (2001). Final results from the hubble space telescope key project to measure the hubble constant. *The Astrophysical Journal*, 553(1):47.
- Gamow, G. (1948). The evolution of the universe. *Nature*, 162(4122):680–682.
- Hamden, E. T., Greer, F., Hoenk, M. E., Blacksberg, J., Dickie, M. R., Nikzad, S., Martin, D. C., and Schiminovich, D. (2011). Uv anti-reflection coatings for use in silicon detector design. *arXiv e-prints*.
- Hatziminaoglou, E. (2009). Virtual observatory: Science capabilities and scientific results. *arXiv e-prints*.
- Hubble, E. (1925). Cepheids in spiral nebulae. *The Observatory*, 48:139–142.
- Hubble, E. (1929). A relation between distance and radial velocity among extra-galactic nebulae. *Proceedings of the national academy of sciences*, 15(3):168–173.
- Johnson, H. L. and Morgan, W. W. (1953). Fundamental stellar photometry for standards of spectral type on the revised system of the yerkes spectral atlas. *The Astrophysical Journal*, 117(3):313–352.
- Kippenhahn, R. and Weigert, A. (1990). *Stellar Structure and Evolution*. Astronomy and Astrophysics Library. Springer-Verlag.
- Krelowski, J. and Papaj, J. (1993). The interstellar extinction curve. *Publications of the Astronomical Society of the Pacific*, 105(693):1209.
- Leavitt, H. S. (1908). 1777 variables in the magellanic clouds. *Annals of Harvard College Observatory*, 60(4):87–103.
- Leavitt, H. S. and Pickering, E. C. (1912). Periods of 25 variable stars in the small magellanic cloud. *Harvard College Observatory Circular, vol. 173, pp. 1-3*, 173:1–3.
- Lemaître, A. G. (1931). Contributions to a british association discussion on the evolution of the universe. *Nature*, 128(3234):704–706.
- Lindgren, L. e. a. (2018). Gaia data release 2: The astrometric solution. *A&A*, 616:A2.

- Madore, B. (1982). The period-luminosity relation. iv-intrinsic relations and reddenings for the large magellanic cloud cepheids. *The astrophysical journal*, 253:575–579.
- Madore, B. F., Freedman, W. L., and Moak, S. (2017). A method for improving galactic cepheid reddenings and distances. *The Astrophysical Journal*, 842(1):42.
- Navarro, J. F. and White, S. D. M. (1993). Simulations of dissipative galaxy formation in hierarchical clustering models. *MNRAS*, 265:271–287.
- Payne-Gaposchkin, C. H. (1925). *Stellar atmospheres: A contribution to the observational study of high temperature in the reversing layers of stars*. PhD thesis, Radcliffe College, United States.
- Penzias, A. A. and Wilson, R. W. (1965). A Measurement of Excess Antenna Temperature at 4080 Mc/s. *The Astrophysical Journal*, 142:419–421.
- Phillips, M. M. (1993). The absolute magnitudes of type ia supernovae. *The Astrophysical Journal*, 413:L105–L108.
- Pillepich, A., Nelson, D., et al. (2025). Introducing the tng-cluster simulation: overview and physical properties of the gaseous intracluster medium. *Astronomy & Astrophysics*. TNG-Cluster is a large-scale cosmological magnetohydrodynamical simulation of galaxy clusters within the IllustrisTNG framework.
- Planck, M. (1900). On the theory of the energy distribution law of the normal spectrum. *Verh. Deut. Phys. Ges*, 2(237):237–245.
- Planck Collaboration, Aghanim, N., Akrami, Y., Ashdown, M., Arroja, F., and ... (2020). Planck 2018 results. vi. cosmological parameters. *Astronomy & Astrophysics*, 641.
- Pogson, N. R. (1856). Magnitudes of thirty-six of the minor planets for the first day of each month of the year 1857. *Monthly Notices of the Royal Astronomical Society*, 17:12–15.
- Riess, A. G. and Filippenko, A. V. (1998). Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant. *Astrophysical journal*, 116(3):1009–1038.
- Riess, A. G., Yuan, W., Macri, L. M., Scolnic, D., Brout, D., Casertano, S., Jones, D. O., Murakami, Y., Anand, G. S., Breuval, L., Brink, T. G., Filippenko, A. V., Jha, S. C., Kenworthy, W. D., Mackenty, J., Stahl, B. E., and Zheng, W. (2022).

- A comprehensive measurement of the local value of the hubble constant with $1 \text{ km s}^{-1} \text{ mpc}^{-1}$ uncertainty from the hubble space telescope and the sh0es team. *The Astrophysical Journal Letters*, 934(1):L7.
- Roth, M. M. (2023). A brief history of image sensors in the optical. *Astronomische Nachrichten*, 344(8–9):–.
- Sandage, A. (1958). Current problems in the extragalactic distance scale. *The Astrophysical Journal*, 127:513.
- Sandage, A., Tammann, G., and Reindl, B. (2004). New period-luminosity and period-color relations of classical cepheids-ii. cepheids in lmc. *Astronomy & Astrophysics*, 424(1):43–71.
- Schwarzschild, K. (1900). Über das gesetz der energieverteilung im spektrum der fixsterne. *Astronomische Nachrichten*, 152:1–10. English title: “On the Law of Energy Distribution in the Spectra of Fixed Stars”.
- Storm, J., Gieren, W., Fouque, P., Barnes, T. G., Pietrzyński, G., Nardetto, N., Weber, M., Granzer, T., and Strassmeier, K. (2011). Calibrating the cepheid period-luminosity relation from the infrared surface brightness technique-i. the p-factor, the milky way relations, and a universal k-band relation. *Astronomy & Astrophysics*, 534:A94.
- Svolopoulos, S. N. (1960). Six-color photometry of ten classical Cepheids. , 65:473–480.
- Taylor, M. B. (2005). Topcat & stil: Starlink table/votable processing software. *Astronomical Data Analysis Software and Systems XIV*, 347:29.
- Welch, D. L. (1994). Near-Infrared Variants of the Barnes-Evans Method for Finding Cepheid Distances Calibrated With High-Precision Angular Diameters. , 108:1421.
- Wesselink, A. J. (1946). The observations of brightness, colour and radial velocity of δ Cephei and the pulsation hypothesis (Errata: 10 258, 310). , 10:91.
- Whiteoak, J. B. (1966). The Wavelength Dependence of Interstellar Extinction. *The Astrophysical Journal*, 144:305.
- Zöllner, J. C. F. (1861). *Photometrische Untersuchungen*. Wilhelm Engelmann, Leipzig.